

Experiment Tracking

Important concepts

- ML experiment: the process of building an ML model
- Experiment run: each trial in an ML experiment
- Run artifact: any file that is associated with an ML run
- Experiment metadata: information about an ML experiment like the source code used, the name of the user, etc.

What's experiment tracking?

Experiment tracking is the process of keeping track of all the **relevant information** from an **ML experiment**, which includes:

- Source code
- Environment
- Data
- Model
- Hyperparameters
- Metrics
- ...

Why is experiment tracking so important?

In general, because of these 3 main reasons:

- Reproducibility
- Organization
- Optimization

Tracking experiments in spreadsheets

Why is not enough?

- Error prone
- No standard format
- Visibility & Collaboration

[illegible]

MLflow

Definition: *“An open source platform for the machine learning lifecycle”* *

In practice, it's just a Python package that can be installed with pip, and it contains four main modules:

- Tracking
- Models
- Model Registry
- Projects

* Official MLflow documentation: <https://mlflow.org/>

Tracking experiments with MLflow

The MLflow Tracking module allows you to organize your experiments into runs, and to keep track of:

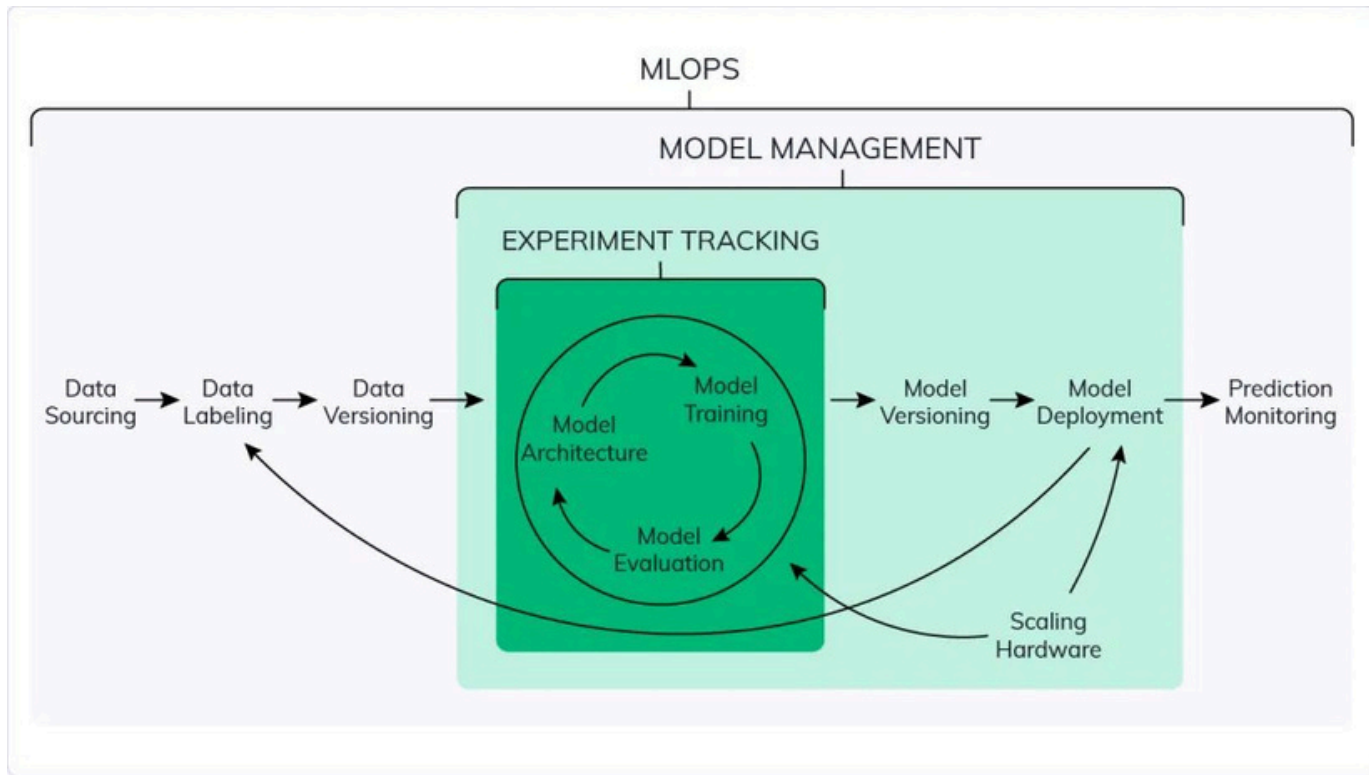
- Parameters
- Metrics
- Metadata
- Artifacts
- Models

Along with this information, MLflow automatically logs extra information about the run:

- Source code
- Version of the code (git commit)
- Start and end time
- Author

Model Management

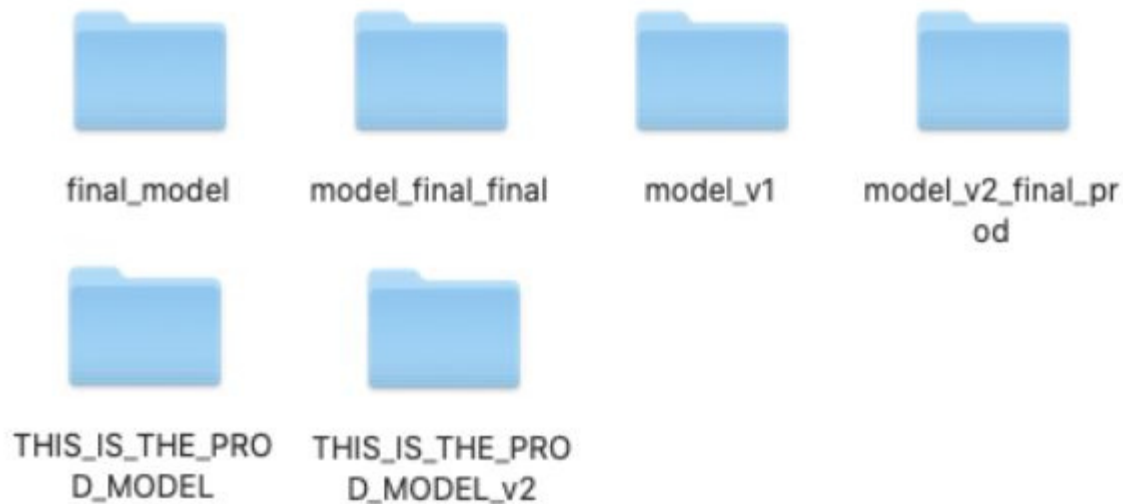
Machine Learning Lifecycle



Model management

What's wrong with this?

- Error prone
- No versioning
- No model lineage



Logging models in MLflow

Two options:

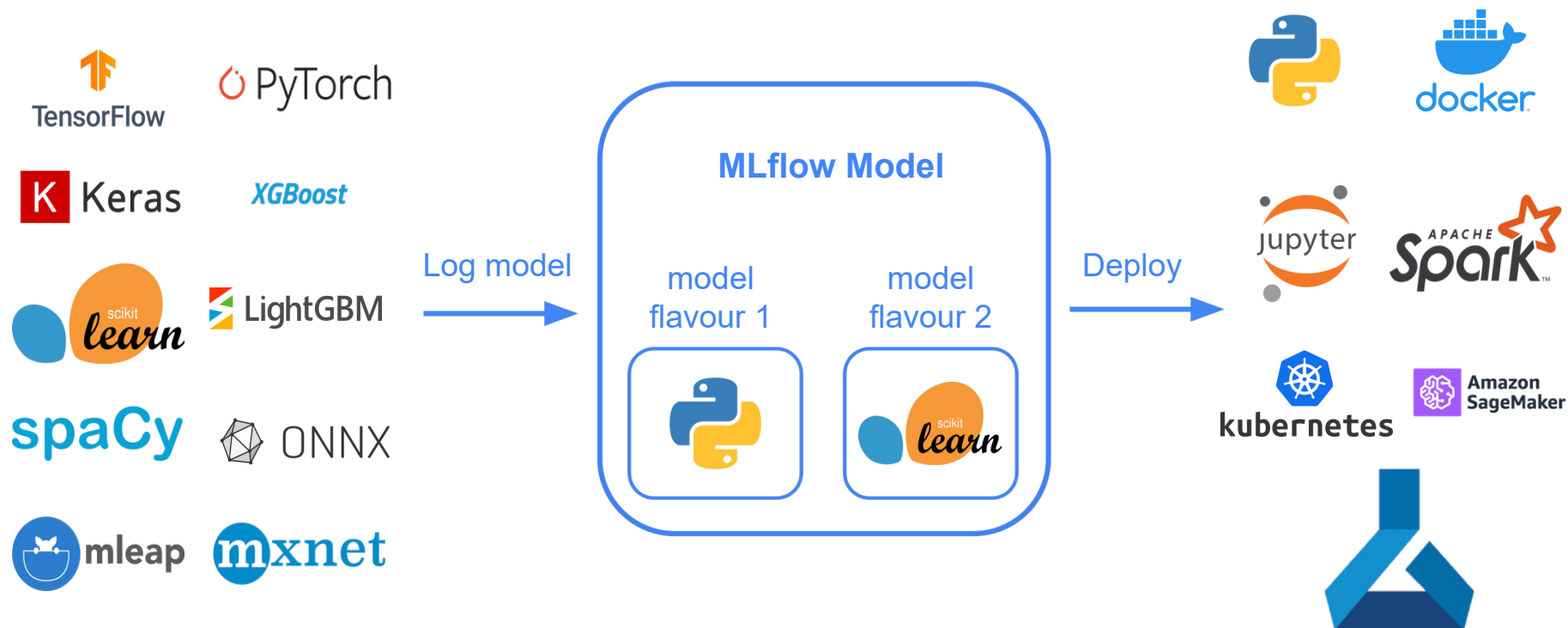
- Log model as an artifact

```
mlflow.log_artifact("mymodel", artifact_path="models/")
```

- Log model using the method `log_model`

```
mlflow.<framework>.log_model(model, artifact_path="models/")
```

MLflow Model Format



Model Registry

Motivation

New model 🔍 Inbox x



Alexey Grigorev

to me ▾

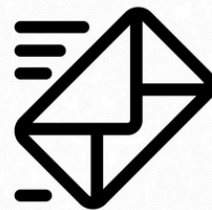
Hi Cristian,

Here's the new model. Can you please deploy it to prod?

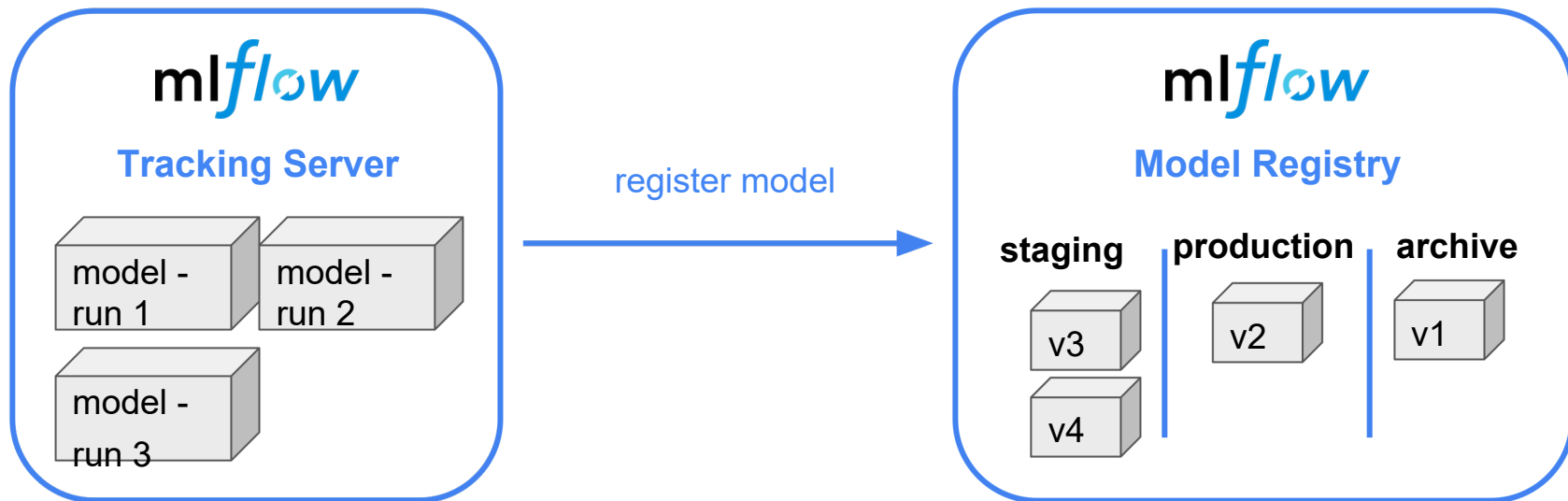
Thanks



category-classifier-...



Model Registry



MlflowClient Class

- A client of ...
 - an MLflow Tracking Server that creates and manages experiments and runs.
 - an MLflow Registry Server that creates and manages registered models and model versions.
- To instantiate it we need to pass a tracking URI and/or a registry URI:

```
from mlflow.tracking import MlflowClient  
  
client = MlflowClient(tracking_uri="sqlite:///mlflow.db")
```


Model management in MLflow

The Model Registry component is a centralized model store, set of APIs, and a UI, to collaboratively manage the full lifecycle of an MLflow Model.

It provides:

- Model lineage,
- Model versioning,
- Stage transitions, and
- Annotations

MLflow in Practice

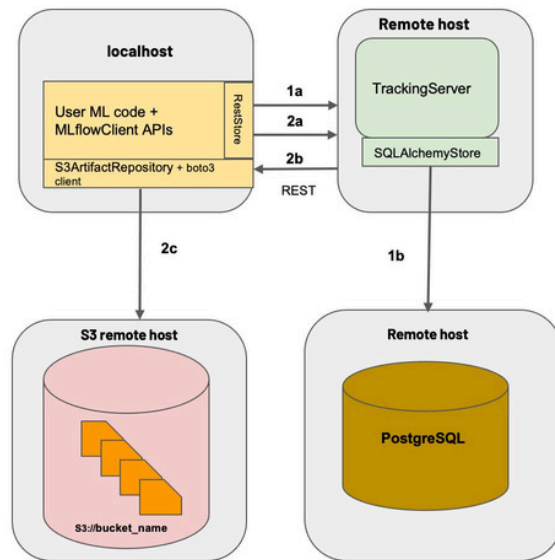
Different scenarios for running MLflow

Let's consider these three scenarios:

- A single data scientist participating in an ML competition
- A cross-functional team with one data scientist working on an ML model
- Multiple data scientists working on multiple ML models

Configuring MLflow

- Backend store
 - local filesystem
 - SQLAlchemycompatible DB (e.g. SQLite)
- Artifacts store
 - local filesystem
 - remote(e.g.s3 bucket)
- Tracking server
 - no tracking server
 - localhost
 - remote

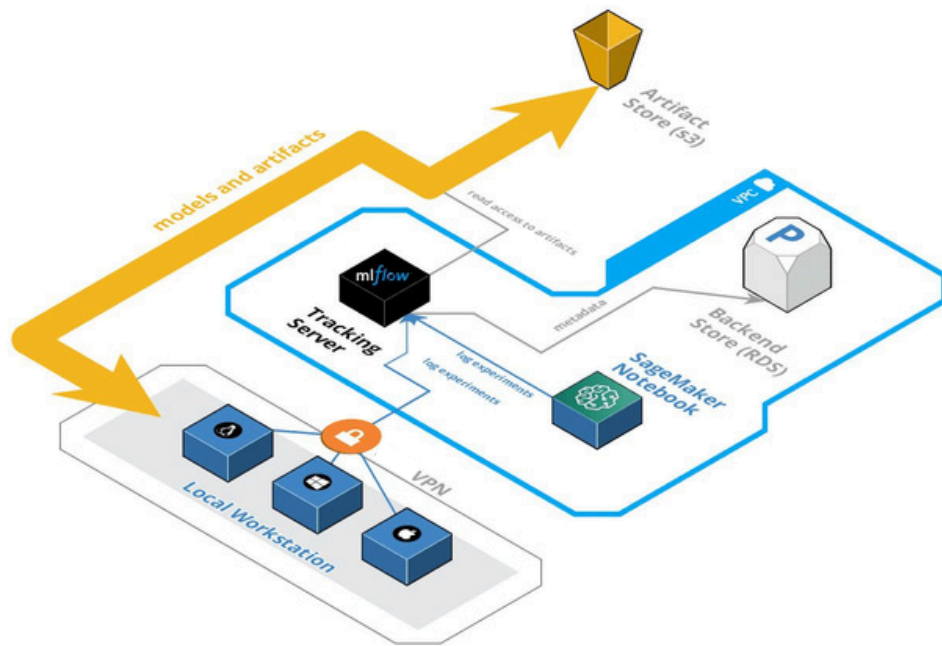


Remote tracking server

The tracking server can be easily deployed to the cloud

Some benefits:

- Share experiments with other data scientists
- Collaborate with others to build and deploy models
- Give more visibility of the data science efforts



MLflow limitations (and when not to use it)

- **Authentication & Users:** The open source version of MLflow doesn't provide any sort of authentication
- **Data versioning:** to ensure full reproducibility we need to version the data used to train the model. MLflow doesn't provide a built-in solution for that but there are a few ways to deal with this limitation
- **Model/Data Monitoring & Alerting:** this is outside of the scope of MLflow and currently there are more suitable tools for doing this

MLflow alternatives

There are some paid alternatives to MLflow:

- [Neptune](#)
- [Comet](#)
- [Weights & Biases](#)
- and [many more](#)

